INCH-POUND
MIL-DTL-3866J
06 November 2020
SUPERSEDING
MIL-DTL-3866H
13 December 2005

DETAIL SPECIFICATION

GLOVES, CLOTH, COTTON, KNITTED, LIGHTWEIGHT

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for lightweight, knitted, cotton, cloth, gloves.

1.2 Classification. This specification covers the following types and sizes.

1.2.1 Types.

Type I - Film handlers
Type II - Inserts

1.2.2 Sizes.

Schedule of sizes: Small (S), Medium (M)

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AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 - Thread, Cotton

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of these documents are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM20 - Test Method for Fiber Analysis: Qualitative

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(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D3776/D3776M	- Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	- Standard Test Method for Bursting Strength of Textiles – Constant – Rate – of – Traverse (CRT) Ball Burst Test
ASTM D6193	- Standard Practice for Stitches and Seams
ASTM D8007	- Standard Test Method for Wale and Course Count of Weft Knitted Fabrics

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to inspection in accordance with 4.2.

3.2 Guide samples. Guide samples, when furnished, shall be used solely for guidance and information purpose to the contract (see 6.3). Variations from this document may appear in the sample in which case this document shall govern.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.4 Design. The glove shall be an ambidextrous design with an unfinished wrist without cuff as shown on Figure 1. Type I gloves shall be inseam constructed and Type II gloves shall be outseam constructed.

3.5 Figure. Figure 1 is furnished for information purposes only. If there are any inconsistencies between the specification and the figure, the specification shall govern.

3.6 Materials.

3.6.1 Fabric. The fabric for the gloves shall be a plain (jersey) circular knit fabric made with single cotton yarns. The finished fabric shall be scoured and bleached. The use of optical brighteners is prohibited. The finished fabric shall conform to the physical requirements in Table I when tested as specified in 4.4.1.

TABLE I. Physical requirements (all types).

Characteristic	Requirements
Knit type	3.6.1
Fiber identification	3.6.1
Weight, oz./sq.yd. (minimum)	3.4
Yarns per inch (minimum) Wales Courses	 28 26
Bursting strength, lbs. (minimum)	50
Optical brighteners	None

3.6.2 Color. The color of the knitted fabric shall be bleached white.

3.6.3 Thread. The cotton thread for seaming and stitching shall be white and shall conform to Type I, Tex 40, 3 ply and Tex 30, 2 ply of A-A-52094.

3.7 Label. Each glove shall have a combination identification and size label conforming to Type III or IV, Class 4 of MIL-DTL-32075, except that the size label shall be on the bottom line and not vertically. Labels shall be centered between side seams of gloves, about 1/2-inch from unfinished edge of wrist. The label for Type I gloves shall be on the inside and for Type II gloves it shall be on the outside.

3.8 Patterns. Standard patterns (size medium) will be furnished by the Government. The Government pattern shall not be altered in any way and shall be used to create the contractor's working patterns. Minor pattern modifications are permitted to accommodate manufacturing procedures; however, the design and finished measurements must be maintained. Pattern guidelines shall not be modified. The pattern parts and cutting quantities shall be as specified in Table II.

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TABLE II. List of pattern parts (Type I and II).

Nomenclature	Cutting quantity per pair
Back	2
Front	2

3.9 Construction.

3.9.1 Manufacturing requirements. The gloves shall be inseam or outseam sewn, as applicable, and shall be made by joining front and back by stitching 1/8 to 3/16-inch from raw edge.

3.9.2 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to ASTM D6193. The stitch type shall be 301 or 401 and the seam and stitching type shall be Ssa-1 for Type I and II gloves. When two (2) or more methods, seams, or stitches are given for the same part of an operation, any one of them may be used. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur.

3.9.2.1 Stitches per inch. The stitches per inch for the gloves shall be 8-10.

3.9.2.2 Type 301 stitching. Ends of 301 stitching shall be backstitched or overstitched 1/4-inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so there will be no loose stitching resulting in a loose bobbin or top thread or no excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the sewn materials.

3.9.2.2.1 Repairs of type 301 stitch. Repairs of type 301stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching.
b. Thread breaks or two (2) or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1-inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and re-stitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

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3.9.2.3 Type 401 stitching. As an alternate gloves can be sewn using Type 401 stitch. Ends of 401 stitching shall be backstitched or overstitched 1/4-inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so there will be no loose stitching resulting in a loose bobbin or top thread or no excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the sewn materials. All repairs shall be made in accordance with 3.9.2.2.1 a. and b. Repairs to Type 401 stitching may be accomplished by use of stitch Type 301.

3.9.3 Finishing. Type I gloves shall be turned right side out and form or lay off. Type II gloves shall not be turned, but shall be laid out flat.

3.10 Finished measurements. The finished measurements for the glove shall be as specified in Table III. The measurements shall be made as indicated in Figure 1, except Type II gloves shall be measured from stitch line to top of glove and from stitch line to stitch line as applicable.

TABLE III. Finished measurements of gloves (inches).

Size	Total length "A" 1/	Hand width "B" 1/	Width across bottom (wrist opening) "C" 1/
Small	14 1/2	3 1/2	3 1/4
Medium	15	4	3 3/4
Tolerance	±1/2	±1/4	±1/8

1/ As located in Figure 1.

3.11 Toxicity. The finished textile materials shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.12 Workmanship. The finished gloves shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.4, shall not exceed the criteria specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color, finished measurements listed in Table III, finished defects listed in 4.4 and tested for the characteristics as specified in 4.4.1.

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4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.4.1 through 4.5 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 End item visual examination. The finished gloves shall be examined for the defects listed in Table IV. The lot size shall be expressed in units of gloves. The sample unit shall be one pair of gloves and selection shall be by pairs. Defects for pairing shall be classified as a single defect.

TABLE IV. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Pairing	- Not matched (i.e. not the same size)	101	
Cleanness	- Spots or stains clearly noticeable affecting appearance - Rancid or other objectionable odor		201 202
Color	- Other than bleached white	102	
Cutting	- Any component not cut on straight line and not cut with length direction of component in wales direction of fabric	103	
Finish	- Thread ends not trimmed - Surplus material on seams not evenly trimmed		203 204
Material	- Any hole larger than 3/32-inch - Three (3) or more holes - Two (2) holes 3/32-inch or less located closer than 3-inches to each other NOTE: Two (2) holes 3/32-inch or less are not to be scored as a defect if the holes are 3-inches or more apart. All holes shall be measured across opening in longest dimension. - Run or dropped stitch - Any clearly noticeable unevenness of fabric - Needle chews, slubby yarn, any cut, tear or burn - Any mend	104 105 106 107 108	 205 206
Fingers and thumb	- Any finger or thumb twisted, pleated, or poorly shaped		207

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TABLE IV. End item visual defects. - Continued

Examine	Defect	Classification	
		Major	Minor
Seams and stitching	- Any open seam - Repair of seam not as specified NOTE: A seam shall be classified as open when one (1) or more stitches joining a seam are broken or when two (2) or more consecutive skipped or run-off stitches occur or raw edge is not caught in stitching	109	 208
Raw edge (except on wrist) Type I only	- More than 1/4-inch on outside of glove		209
Seam and stitch type	- Wrong seam or stitch type		210
Construction	- Type I glove not inseam sewn - Type II glove not outseam sewn		211 212
Stitches per inch	- Less than eight (8) stitches per inch - More than ten (10) stitches per inch NOTE: To be scored only when the condition exists on the major portion of a seam	110	 213
Stitch tension (301 or 401 stitch type)	- Loose tension resulting in a loosely secured seam - Tight tension resulting in clearly noticeable puckering along seam (stitches break when normal strain is applied to seam or stitching)	111 112	
Ends of stitching	- Ends of stitching produced with 301 or 401 stitch type, backstitched less than 1/4-inch		214
Inseam	- Stitching less than 1/8-inch or more than 3/16-inch from edge		215
Label	- Missing, incorrect, or illegible - Misplaced	113	 216

4.4.1 Material/end item testing. The finished knitted fabric shall be tested for the characteristics listed in Table V. The methods of testing as specified wherever applicable and as listed in Table V shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be one (1) yard full width of the finished fabric for all physical tests. The lot shall be unacceptable if one (1) or more sample units or the lot average fail to meet any requirement specified. The sample size shall be in accordance with the following:

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Lot size (units)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE V. Material/component tests.

Characteristic	Requirement reference	Test method
Knit type	3.6.1	Visual
Fiber identification	3.6.1	AATCC TM20 (see 6.4)
Weight	Table I	ASTM D3776/D3776M, Option C
Yarns per inch	Table I	ASTM D8007
Bursting strength	Table I	ASTM D3787
Optical brighteners	Table I	AATCC EP9, Option C (see 6.5), 1/, 2/

1/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

2/ Sample shall be viewed under UV illumination. .The appearance of a bright blue hue of fluorescent shall be considered a failure.

4.5 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the textile material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.11) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Type I gloves are intended for use in handling film and other similar items. Type II gloves are intended for use as inserts for Gloves Set, Chemical Protective, MIL-DTL-43976.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When a first article is required (see 3.1, 4.2).
e. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
f. When toxicity testing is required (see 3.11, 4.5)
g. Inspection conditions (see 4.3.1)
h. Packaging requirements (see 5.1).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (±5°) angle relative to the horizontal. The observer will view the specimen at a 90° angle, relative to the plane of the specimens."

6.6 Subject term (key word) listing.

Environmental
Hand wear
Hand protection
Waterproof

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

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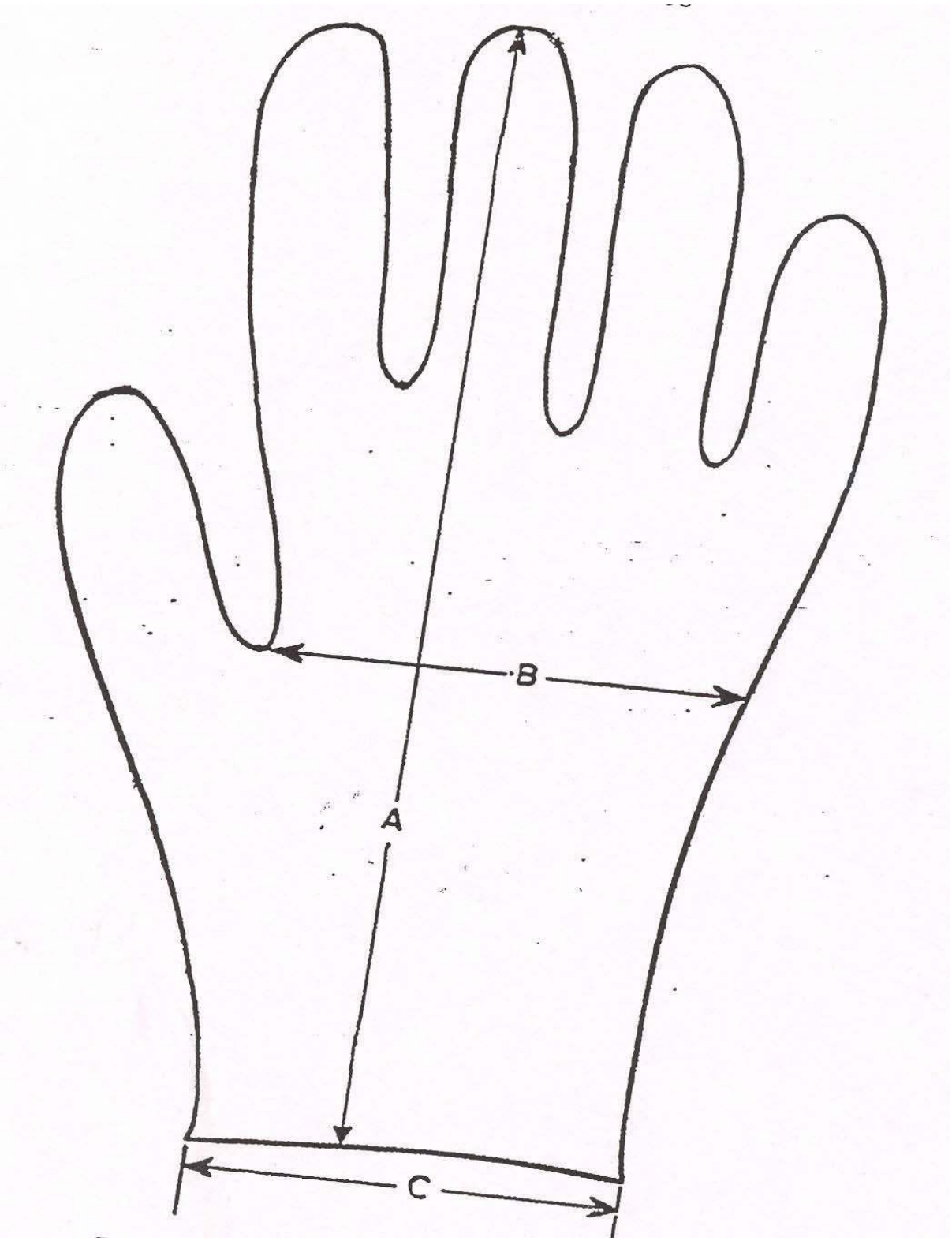


FIGURE 1. Gloves, Men's, Cloth, Knitted, Lightweight.

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Navy - NU
Air Force - 11

Review Activities:
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